STA630

Short Lectures 19-41 by

WANIA HAREEM

Follow this link to join my WhatsApp community:

https://chat.whatsapp.com/GnlgNXt89RPCDqheuikAcB

LESSON # 19 : Research Design

A research design is a master plan outlining how to collect and analyze data. It is a strategy that guides the entire research project. It involves making several decisions based on available options, including the study's purpose, unit of analysis, time dimension, mode of observation, sampling design, observation tools, data processing, and data analysis.

1. Purpose of the Study

Research can be exploratory, descriptive, or explanatory:

- **Descriptive**: Focuses on who, what, where, when, or how much.

- **Explanatory**: Focuses on why and how one variable affects another.

Methodological rigor increases from exploratory to explanatory studies, involving more complex data collection and analysis.

2. Unit of Analysis

The unit of analysis is the level of aggregation of data:

- **Individual**: Studying single persons.

- **Dyads**: Studying interactions between two people (e.g., husband-wife).

- **Groups**: Studying group effectiveness by aggregating individual data.

- **Organizations**: Studying entire departments or organizations.

3. Time Dimension

Observations can be made at one time or over a long period:

Cross-Sectional Studies: Conducted once, representing a snapshot.

- **Longitudinal Studies:** Repeated over time, tracking changes.

4. Researcher Control of Variables

Researchers may or may not control variables:

- **Experimental Design**: Manipulates variables to study effects (can be contrived or non-contrived).
- **Ex Post Facto Design**: Observes existing conditions without manipulation.

5. Mode of Observation

Data collection methods depend on the study's nature:

- Survey
- Experiment
- Content Analysis
- Field Observation
- Case Study
- Focus Group Discussion

6. Sampling Design

Sampling involves selecting a subset of the population:

- A population element is the unit of analysis.
- Sampling aims to draw conclusions about the entire population.

7. Observation Tools

Common tools include:

- Questionnaire
- Interview Schedule
- Interview Guide
- Checklist

The choice of tool depends on the study's logic and appropriateness.

8. Field Data Collection

The procedure for field operations involves:

- Deciding how data will be collected.

- Assigning responsibility for data collection.

- Training field personnel.

Ensuring data quality control.

9. Data Processing and Data Analysis

Researchers outline how data will be processed and analyzed:

- Specify whether data processing will be manual or mechanical.

- Explain how qualitative data will be quantified.

- Describe the use of statistics and the inferences to be drawn.

Survey Research: An Overview

Surveys gather primary data by asking respondents questions using questionnaires or interviews. Steps in conducting a survey include:

1. Develop hypotheses, choose the survey type, write questions, and design the layout.

2. Plan data recording methods and pilot test the survey instrument.

3. Determine the target population, obtain a sampling frame, decide on sample size, and select the sample.

4. Locate respondents, conduct interviews, and record data accurately.

5. Enter data into computers, recheck all data, and perform statistical analysis.

6. Describe methods and findings in a research report and present them for critique and evaluation.

Lesson 20: Survey Research

Research design can be classified by how primary data is gathered: by observing or by communicating with people. Surveying involves asking people questions and recording their responses. Surveys are versatile for collecting primary data.

Human Interactive Media and Electronic Interactive Media

- Human Interactive Media: Involves face-to-face or telephone conversations. Traditional interviews fall into this category.

- Electronic Interactive Media: Uses digital technology like the Internet to interact with respondents, allowing two-way communication.

Non-Interactive Media

- Non-Interactive Media: Includes mail questionnaires that don't allow immediate feedback or interaction.

Choosing a Communication Media

Once surveying is chosen for data collection, various methods can be used, such as personal interviews, telephone, mail, or computer-based surveys.

Personal Interviewing

A personal interview is a two-way conversation where an interviewer gathers information from a respondent. The interviewer controls the discussion and typically interacts with strangers.

Advantages of Personal Interviewing:

1. **Opportunity for Feedback**: Interviewers can reassure and clarify questions for respondents.

2. **Probing Complex questions**: Allows follow-up questions for more detailed answers.

3. **Length of Interview**: Can be longer than other survey methods, sometimes over an hour.

4. **High Completion Rate**: Face-to-face interaction increases the likelihood of completing the survey.

5. **Props and Visual Aids**: Allows showing samples or visual aids to respondents.

6. **High Participation Rate**: More people are willing to participate in face-to-face interviews.

7. **Observation of Non-Verbal Behavior**: Interviewers can note facial expressions and body

language.

8. **Non-Literates can participate**; Illiterate respondents can also take part since theydon't need to read or write.

9. **Interviewer can Prescreen Respondent:** Ensures the respondent meets the surveycriteria.

10. **CAPI (Computer Assisted Personal Interviewing:** Responses can be entered into acomputer to reduce errors and costs.

Disadvantages of Personal Interviewing:

1. **High Cost:** More expensive than mail, internet, and telephone surveys.

2. **Scarcity of Highly Trained Interviewers:** Difficult to find qualified interviewers forlarge studies.

3. **Lack of Anonymity**: Respondents might be reluctant to share confidential information.

4. **Callbacks:** Labor-intensive and costly when respondents are not available on the first attempt.

5. **Interviewer Influence**: Respondents' answers may be influenced by the interviewer's characteristics.

6. **Interviewer Bias**: Interviewer's personal biases can affect the data.

7. **No Opportunity to Consult:** Respondents might not have access to records during the interview.

8. **Less Standardized Wording**: Interviewers might not use the exact wording, potentially biasing the data.

9. **Limitations in Respondents' Availability and Accessibility**: Some people, like executives, may be hard to reach.

10. **Some Neighborhoods are Difficult to Visit**: Security concerns may prevent access.

Door to Door Interviews

- Conducted at the respondent's home or workplace. Provides a more representativesample but may exclude people in secure buildings or busy executives.

Intercept Interviews in Malls and Other High-Traffic Areas:

- **Mall Intercept Interviews**: Conducted in shopping malls, low cost, but high refusal rates. Not suitable for representative samples of the total population but canbe useful for specific demographics.

Summary: Personal interviews are a valuable method for collecting survey data due to their ability to gather detailed responses and use visual aids, despite being more costly and labor- intensive. Electronic and non-interactive media also play significant roles in modern surveyresearch.

LESSON 21

INTERCEPT INTERVIEWS IN MALLS AND OTHER HIGH-TRAFFIC AREAS

Mall intercept interviews are personal interviews conducted in shopping malls. Interviewers stop shoppers at central points or entrances to ask questions. These interviews are low-cost since there's no need for travel. Many interviews can be done quickly, but refusal rates are high because people are often in a hurry. Researchers shouldnote that mall intercepts may not provide a representative sample of the total population. They are suitable when demographic factors don't affect the survey results or when targeting a specific group, like parents of bike-riding children.

Telephone Interviewing:
Telephone interviewing is common in survey research and provides data quality similar to personal interviews. Respondents might share more personal information over the phone.Telephone surveys can provide representative samples and are usually conducted fromcentral locations. Advances in technology, such as Computer-Assisted Telephone Interviewing (CATI), allow responses to be directly entered into a computer.

Strengths of Telephone Interviewing

1. **High Speed**: Data collection is fast, allowing many interviews in a short time.
2. **Saves Cost**: Cheaper than personal interviews.
3. **Callbacks**: Easier and less expensive than personal callbacks.
4. **Random Digit Dialing**: Can be used for random sampling.
5. **Geographic Coverage:** Covers larger areas without increasing costs.
6. **Fewer Interviewers Needed**: Uses skilled interviewers efficiently.
7. **Reduced Bias**: Less interviewer bias.
8. **Access to Hard-to-Reach Respondents**: Easier to reach reluctant respondents.
9. **CATI**: Reduces errors and costs.

Weaknesses of Telephone Interviewing

1. **No Face-to-Face Contact**: Less personal, which can be a drawback.
2. **Lower Response Rate**: Some refuse to participate.

3. **No Visual Aids**: Can't use visual materials.
4. **Limited Duration:** Interviews should be short (about 10 minutes).
5. **Unlisted Numbers:** Many numbers are not accessible.
6. **Less Engagement**: Participants might be less thorough.
7. **Distractions**: Multiple phones can distract.

Self-Administered Questionnaires
These are common in everyday life, such as service evaluations. They can be delivered bymail, fax, courier, or online.

Mail Questionnaire: A self-administered questionnaire sent through mail.

Advantages of Mail Questionnaire:

1. **Geographic Flexibility**: Can reach widespread areas cheaply.
2. **Sample Accessibility**: Can contact hard-to-reach individuals.
3. **Saves Time**: Quick distribution and assessment.
4. **Saves Cost**: Cheaper than personal and phone interviews.
5. **Respondent Convenience**: Can be filled out at the respondent's convenience.
6. **Anonymity**: Provides a sense of anonymity, encouraging honest responses.
7. **Standardized Questions**: Highly structured questions.

Disadvantages of Mail Questionnaire

1. **Low Response Rate**: Few returned questionnaires.
2. **Low Completion Rate**: Some questions may be left unanswered.
3. **Increases Cost:** Follow-up reminders add to the cost.
4. **Interviewer's Absence**: No help for clarifications.
5. **No Control on question Order**: Respondents see all questions at once.
6. **Length Limits:** Long questionnaires are often avoided.
7. **No Control Over Environment:** Uncertain who actually fills it out.
8. **No Non-Verbal Cues:** Cannot observe respondent behavior.

9. **Non-Literates Excluded:** Only literate respondents can participate.

Increasing Response Rate:

Here are some guidelines to increase the response rate. The response rate is the number of completed questionnaires divided by the total number of people contacted.

Cover Letter: A cover letter with the questionnaire can encourage completion. Explain the study's importance, sponsor, how the respondent was chosen, and assure anonymity.A personalized letter makes the respondent feel important. An individually typed letter on letterhead increases response rates.

Money Helps: Offering money or incentives like pens or lottery tickets can motivaterespondents. Monetary incentives are the most effective and work for all income levels.

Interesting questions: Add some interesting questions at the beginning of thequestionnaire to engage respondents and encourage cooperation.

LESSON 22

SELF ADMINISTERED QUESTIONNAIRES (CONTINUED)

Increasing Response Rate:
Here are some guidelines to increase the response rate. The response rate is the number ofcompleted questionnaires divided by the total number of people contacted.

Cover Letter: A cover letter with the questionnaire can encourage completion. Explain the study's importance, sponsor, how the respondent was chosen, and assure anonymity.A personalized letter makes the respondent feel important. An individually typed letter on letterhead increases response rates.

Money Helps: Offering money or incentives like pens or lottery tickets can motivaterespondents. Monetary incentives are the most effective and work for all income levels.

Interesting questions: Add some interesting questions at the beginning of thequestionnaire to engage respondents and encourage cooperation.

Follow-Ups: Follow-up reminders, including duplicate questionnaires or simple reminders, can increase response rates. Multiple contacts improve the chances of responses.

Preliminary Notification: Inform respondents in advance, by letter or phone, that theywill receive a questionnaire. This helps increase response rates.

Survey Sponsorship: Sponsorship by well-known and respected organizations canincrease response rates due to their goodwill.

Return Envelopes: Include a stamped, self-addressed envelope to make it easier forrespondents to return the questionnaire.

Postage: Using first-class or expedited delivery can increase response rates.

Personalization: Personalizing mailings may help, but results vary. Typed letters can besomewhat effective.

Size, Reproduction, and Color: Use A-4 size paper and avoid folding it. Attractive printing and different colors for different parts of the questionnaire can increase interest.

Combining multiple techniques may be needed to significantly boost response rates.

E-Mail Surveys: E-mail questionnaires are fast, cost-effective, and flexible. They work well for internal surveys or frequent e-mail users. Assure respondents of confidentiality, ase-mail may feel less secure.

Internet Surveys: These are self-administered on websites. Respondents answerquestions displayed on the screen.

Advantages of Internet Surveys:

- **Speed and Cost Effectiveness**: Internet surveys are quick and cost less, reaching a large audience easily.
- **Visual Appeal and Interactivity**: They can be interactive and engaging with visuals and animations.
- **Respondent Participation and Cooperation**: Some respondents actively participate inonline surveys.
- **Accurate Real-Time Data Capture:** Responses are entered directly into the system,allowing real-time data analysis.
- **Callbacks**: Follow-up with non-respondents is easy.
- **Personalized and Flexible questioning**: The survey can adapt based on previousanswers.
- **Respondent Anonymity**: Respondents feel more comfortable providing honestanswers anonymously.

Disadvantages of Internet Surveys:

- **All People Cannot Participate**: Not everyone has internet access or the necessary technology.
- **No Physical Incentive**: Unlike mail surveys, no physical incentives can be provided.

Selecting the Appropriate Survey Research Design:
Choose a method by comparing research objectives with the strengths and weaknesses of each method. There is no "best" survey form. Each has advantages and disadvantages.

Consider factors like the need for an interviewer, respondent interest, cooperation ease, speed, questionnaire complexity, and budget. Combining methods can offer flexibility andmeet project needs.

LESSON 23:TOOLS FOR DATA COLLECTION

TOOLS FOR DATA COLLECTION

There are three main tools for data collection in communication surveys:

1. **Interview Schedule**
2. **Questionnaire**
3. **Interview Guide**

An interview schedule and a questionnaire are both pre-designed lists of questions. In aninterview schedule, the interviewer asks the questions and records the responses. In a Questionnaire, the respondent reads the questions and records their answers. For convenience, "questionnaire" will refer to both types.

An Interview guide is a list of topics to cover during an in-depth interview. Questions aremade up on the spot, usually open-ended, and the sequence may vary.

Guidelines for Questionnaire Design:
A good survey depends on good questions. Designing a questionnaire requires more thancommon sense and good grammar. Questions must be clear, understandable, and relevant.

1. **Questionnaire Relevancy**: Collect only the information needed to solve theproblem. Avoid irrelevant questions.

2. **Questionnaire Accuracy**: Use simple, clear, unbiased words. Avoid jargon, slang, and abbreviations. Ensure questions are understandable to all respondents.

3. **Avoid Ambiguity, Confusion, and Vagueness** make questions specific to avoid confusion. Avoid indefinite words like "often," "usually," and "fair."

4. **Avoid Double-Barreled Questions**: Each question should ask only one thing. Forexample, instead of "Does this company have pension and health insurance benefits?" ask about each separately.

5. **Avoid Leading Questions**: Do not lead respondents to a particular answer. Forexample, instead of "You don't smoke, do you?" ask, "Do you smoke?"

6. **Avoid Loaded Questions:** Avoid questions that suggest a socially desirableanswer or are emotionally charged.

7. **Avoid Burdensome Questions**: Avoid questions that tax memory. For example,asking about feelings or events from childhood may be difficult to answer.

8. **Arrange Questions in Proper Sequence**: Start with interesting, easy questions.Use the funnel technique: start with general questions before moving to specifics.

9. **Use Filter Questions if Needed**: Use filter questions to screen out respondentswho are not qualified to answer subsequent questions.

10. **Layout of the Questionnaire:**

- Ensure good layout and physical attractiveness.
- Number each question and include identifying information.
- Use a cover sheet for administrative use, including date, time, and interviewer details.
- Include clear instructions and print them differently from questions.
- For mail surveys, use a professional cover letter and end with "Thank you for your participation."

Lesson 24

PILOT TESTING OF THE QUESTIONNAIRE

Pilot testing, also known as pre-testing, refers to a small-scale trial run of a particular component, specifically the questionnaire in this context. Pre-testing is an establishedpractice for identifying errors and providing extra training for the research team.
Professionals with extensive experience in studies are more likely to pre-test an instrument than novice researchers, who often underestimate the necessity of the design-test-revise process. Revising questions multiple times is common.

Pilot testing ensures that questions are understood by respondents and identifies problems with wording or measurement. It involves a small number of respondents similarto the final sample to test the questionnaire's appropriateness and comprehension. This can prevent the disaster of using an invalid questionnaire on a larger scale. The main purpose is to identify potential problems with

methods, logistics, and the questionnaireitself.

Administering a questionnaire exactly as planned is often not feasible. For instance, mailing out a questionnaire might take weeks and may not reveal why questions were skipped or misunderstood. Personal interviewers can provide valuable insights into respondents' difficulties with questions.

Aspects to be Evaluated during Pilot Testing

1. Reactions of Respondents:

- **Local Culture Familiarity**: First-hand experience is valuable even for researchersfamiliar with the local culture.
- **Availability of Respondents**: Pre-testing helps find the best time to contact respondents.
- **Acceptability of Questions:** It reveals participants' reactions to the questions, highlighting issues like repetitiveness, sensitivity, and offensive wording.
- **Willingness to Cooperate**: It gauges the likely level of cooperation from respondents.

2. Discovering Errors in the Instrument:

- **Information Reliability**: Ensures the questionnaire provides reliable and analyzabledata.
- **Time Needed**: Indicates the time required for interviews or observations, helping tomodify long questionnaires and estimate average interview times.
- **Format Revisions**: Helps in arranging questions, wording, translation, spacing,answer structuring, coding, and instructions for interviewers.

3. Sampling Procedure Check:

- **Instruction Adherence**: Evaluates if field workers follow sampling instructions andidentifies problems and solutions.
- **Time to Locate Respondents**: Assesses the ease and time required to find respondents, aiding in overall time and budget calculations.

4. Staffing and Research Team Activities:

- **Training Effectiveness**: Pre-testing serves as extra training, evaluating trainingsuccess and identifying additional training needs.
- **Work Output:** Calculates the average output of each fieldworker, determining thenumber needed to complete the work on time and aiding in budget estimates.

- **Team Coordination**: Observes team coordination, addressing any efficiency issues.

- **Logistical Support**: Evaluates the adequacy of logistical support such as transportation, boarding, lodging, guidance, and supervision.

5. Procedure for data processing and analysis can be evaluated:

- **Dummy Tables**: Helps in planning data tabulation and choosing appropriate statistics for interpretation.

Pilot testing is crucial for refining the questionnaire and ensuring the research's success.

LESSON 25: INTERVIEWING

Fieldworkers, including personal interviewers, telephone interviewers, and observers, vary in their activities depending on the data collection method. However, some basic issues apply across all fieldwork settings. This lesson focuses on the interviewing process conducted by personal interviewers.

Who Conducts the Fieldwork?

Data collection in sponsored studies is rarely done by the project designer. Students typically collect their own data, depending on the sample size. The research project reliesheavily on the quality of data collected, so selecting capable fieldworkers is crucial. Field Interviewing Services may be used for door-to-door surveys, telephone interviewing, and other fieldwork. These agencies often employ field supervisors to oversee and train interviewers, edit completed questionnaires, and confirm interview completion.

In-House Training

Fieldworkers must be trained to ensure uniform data collection. Training programs may vary from brief sessions to week-long programs. Training covers how to make initial contact, ask survey questions, probe, record responses, and terminate the interview. Thegoal Is to provide uniform information to all respondents.

The Role of the Interviewer

Survey research interviewing aims to obtain accurate information through a structured conversation. Interviewers build rapport while remaining neutral and objective. They reduce respondents' embarrassment, fear, and suspicion to make them comfortable. Good interviewers monitor the social interaction and content of responses without revealing their opinions.

Stages of an Interview

1. **Making Initial Contact and Securing the Interview:** Interviewers make opening remarks to convince respondents of the survey's importance. For example, "Asslaam-o-Alaykum, my name is [Name] and I am working for a National Survey Company. We are

conducting a survey concerning 'women empowerment.' I wouldlike to get a few of your ideas."

2. **Asking the Questions**: Interviewers ask questions exactly as worded in the questionnaire, read questions slowly, ask them in order, and repeat misunderstoodquestions. They should not take shortcuts even if the task becomes monotonous.

There are five major principles for asking questions:

1. Ask the questions exactly as they are worded in the questionnaire.

2. Read each question very slowly.

3. Ask the question in the order in which they are presented in the questionnaire. 4. Repeat questions that are misunderstood or misinterpreted.

5. Interviewers should record responses under the relevant question, even if answered out of sequence. Instead of skipping these questions, interviewers should say, "We have briefly discussed this, but let me ask you..." This ensures all questions are asked and complete answers are recorded.

Probing: Probing involves encouraging respondents to clarify or expand on theiranswers without suggesting the interviewer's own ideas. Techniques include repeating the question, using an expectant pause, repeating the respondent's reply,and asking neutral questions.

Interviewers have several probing tactics:

1. **Repetition of the question:** Repeating the question can help if the respondent is silent, indicating they may not have understood or decided how to answer.
2. **Expectant pause:** A silent probe with an expectant look can encourage the respondent to elaborate.
3. **Repetition of the respondent's reply:** Repeating the response verbatim may prompt the respondent to expand on their answer.
4. **Neutral questions or comments:** Asking neutral questions can clarify the type of information needed, such as motives or specific meanings of words or phrases.

3. **Recording the Responses**: Responses to closed-ended questions should bechecked off correctly. Open-ended responses should be recorded verbatim, including all relevant information and probes.

4. **Terminating the Interview**: Fieldworkers should not rush to close the interview andshould answer any questions the respondent has about the study. They shouldleave the respondent with a positive feeling and edit the questionnaire in a quietplace, noting details like date, time, and any unusual circumstances.

Principles of Interviewing

- **Integrity and Honesty**: Be honest and ethical in all interactions.
- **Patience and Tact:** Show patience, tact, and courtesy.
- **Accuracy and Detail**: Ensure accurate and detailed recording of responses.
- **Interest and Impartiality**: Show interest without expressing personal opinions.
- **Good Listening:** Listen more than you talk.
- **Confidentiality**: Keep the inquiry and respondents' responses confidential.
- **Respect**: Respect respondents' rights and encourage their cooperation.

Interview Bias

Bias can be introduced by the interviewer, interviewee, or the situation. Interviewer bias includes lack of rapport, unintentional errors, intentional subversion, expectations basedon appearance, failure to probe properly, and influence due to the interviewer's appearance or attitude. Interviewee bias includes providing expected answers, misunderstanding questions, being influenced by the interviewer's appearance, and giving socially desirable answers. Situational bias includes non-participation, varying trust levels,and physical interview settings.

Tips for Interviewing

- Know the culture of the respondents.
- Wear acceptable dress.
- Be pleasant and flexible.
- Carry a letter of authority.
- Establish credibility and rapport.
- Be familiar with the questionnaire.
- Follow question wording and order.
- Record responses exactly.
- Probe for responses.
- Close the interview respectfully and edit the questionnaire promptly.

LESSON 26

SAMPLE AND SAMPLING TERMINOLOGY

Sample: A sample is a part of a larger whole. This whole, called a population, can be anything like a bucket of water, a group of students, or organizations. A population is the total group fromwhich a sample is drawn.

Why Sample?

1. **Saves Cost, Labor, and Time**: Sampling is practical due to budget and time constraints.

2. **Quality Management/Supervision**: Easier to manage and ensure quality with asmaller group.

3. **Accurate and Reliable Results**: Properly selected samples can accurately represent a population.

4. **Sampling May Be the Only Way**: Necessary in destructive testing (e.g., testing firecrackers).

5. **Determine the Period of Study**: Sampling helps avoid the influence of seasonalvariations in long studies.

Sampling Terminology:

- **Element**: The unit about which information is collected (e.g., a person, family, organization).
- **Population**: The total group of elements sharing common characteristics.
- **Target Population:** The specific group of elements relevant to the research project.
- **Sampling**: Using a small number of items from a larger population to make conclusions about thewhole population.
- **Sampling Frame**: The list of elements from which the sample is drawn. It is also called the workingpopulation.
- **Sampling Frame Error**: Occurs when certain sample elements are excluded or the population is not accuratelyrepresented.
- **Sampling Unit:** The element or set of elements considered for selection in some stage of sampling. It caninvolve multiple stages, like selecting neighborhoods, then households, and finally individuals.
- **Observation Unit:** The element from which information is collected. It can differ from the unit of analysis.
- **Parameter**: A summary description of a variable in a population (e.g., the mean income of all

families ina city).

- **Statistic**: A summary description of a variable in a survey sample, used to estimate population parameters.
- **Sampling Error**: The error expected for a given sample, estimated using probability theory.

Summary:

Sampling is crucial for cost-effective and efficient research. Understanding the terminology helps in designing accurate and reliable studies.

LESSON 27

PROBABILITY AND NON-PROBABILITY SAMPLINNG

Probability and Non-Probability Sampling:

-Probability Sampling: Every element in the population has a known, nonzero chanceof being selected. It's used when the sample's representativeness is crucial.

-Non-Probability Sampling: The probability of selecting any particular element isunknown. Selection is arbitrary and relies on personal judgment.

Types of Non-Probability Sampling:

1. **Convenience Sampling:**

- Sampling people who are easiest to reach.
- Examples: Interviewing employees in a nearby area, friends, neighbors, or people on thestreet.
- It's cheap and quick but not very reliable.

2. **Purposive Sampling:**

- The researcher sets criteria for selecting subjects.
- The sample is selected based on these criteria, either by the researcher or someone else.
- Example: Studying students who are regular and active in class discussions.

3. **Quota Sampling:**

- Ensures certain population characteristics are represented.
- The researcher fixes the number of people in various categories.
- Example: Selecting a set number of males and females in different age groups.

4. **Snowball Sampling:**

- Starts with a few cases and expands through referrals.
- Useful for hard-to-identify respondents.
- **Example**: Studying friendship networks among teenagers by asking initial contacts to refertheir friends.

5. **Sequential Sampling**:

- Similar to purposive sampling but stops when new information levels off.
- Continues until no new, diverse information is obtained.
- Example: Interviewing widows until no new insights are gained.

6. Theoretical Sampling:

- Sampling guided by the need for developing a theory.
- Selects cases based on the potential to provide new insights.
- **Example**: Observing a site during different times to see if behavior changes.

Summary

Probability sampling ensures representativeness, while non-probability sampling is used for quick, inexpensive data collection. Different non-probability methods serve variousresearch needs.

LESSON 28

TYPES OF PROBABILITY SAMPLING

Probability Samples

- **Random Processes:** Require more work than nonrandom samples. Specific sampling elements must be identified.
  - Example: In telephone surveys, specific persons must be reached, requiring multiple call attempts.
- **Advantages:**
  - More likely to represent the population.

- Allows statistical calculation of sampling error, defined as the deviation between sample result and population parameter due to random processes.

Simple Random Sample

- **Definition:** The simplest form of random sampling.
  - **Steps:**
    1. Develop an accurate sampling frame.
    2. Select elements using a random procedure.
    3. Locate and include the selected elements in the sample.
  - **Process:** Number all elements, use a list of random numbers to select elements.
  - **Tools:** Random number tables or computer programs.
  - **Note:** Does not guarantee a perfect representation but allows calculation of the probability of inaccuracy.

Systematic Random Sample

- **Definition:** A shortcut for random selection in simple random sampling.
  - **Process:**
    1. Number each element in the sampling frame.
    2. Calculate a sampling interval (K).
    3. Select elements by skipping a set number of elements.
  - **Example:** For a sample size of 300 from a population of 900, the interval is 3.
  - **Note:** Not suitable when elements are in a cycle or pattern.

Stratified Random Sample

- **Definition:** Used when the population is heterogeneous.
  - **Process:**
    1. Stratify the population into homogeneous groups.
    2. Draw a sample using random numbers within each stratum.
  - **Reasons for Use:**
    1. Increase statistical efficiency.
    2. Provide adequate data for subpopulations.
    3. Enable different methods and procedures for different strata.
  - **Types:**
    - Proportionate: Sample size proportional to population size of strata.
    - Disproportionate: Sample size dictated by analytical considerations.

Cluster Sampling

- **Purpose:** Economical sampling while retaining probability sample characteristics.
  - **Process:**

1. Identify clusters with intra-group heterogeneity and inter-group homogeneity.
2. Randomly sample clusters, then elements within selected clusters.
   - **Advantages:** Useful for dispersed populations and reduces costs.

- **Example:** Three-stage sampling: city blocks, households within blocks, individuals within households.
- **Note:** Lower unit costs but higher potential biases.

Double Sampling

- **Definition:** Used when further information is needed from a subset of an already sampled group.
  - **Process:** Initial sample collects preliminary information, and a sub-sample is used for detailed examination.

Appropriate Sample Design

- **Criteria:**
  1. **Degree of Accuracy:** Balance between accuracy and cost/benefit.
  2. **Resources:** Financial and human resource limitations.
  3. **Advance Knowledge of Population:** Availability of lists of population members.
  4. **National vs. Local Project:** Geographic proximity of population elements.
  5. **Need for Statistical Analysis:** Non-probability sampling does not allow statistical projections.

LESSON 29 DATA ANALYSIS

Data Analysis and Processing

Once the data begins to flow in, attention turns to data analysis. If the project has been done correctly, the analysis planning is already done. At the research design stage or at least by the completion of the proposal or the pilot test, decisions should have been made about how to analyze the data.

Editing

Editing is the process of checking and adjusting the data for omissions, legibility, and consistency to ensure the completeness, consistency, and readability of the data to be transferred to data storage.

Field Editing

Field editing is conducted on the same day as the interview to catch technical omissions, check legibility of the handwriting, and clarify responses that are logically or conceptually inconsistent.

In-House Editing

In-house editing rigorously investigates the results of data collection. The in-house editor ensures that inconsistent or contradictory responses are adjusted and that answers will not be a problem for coders and keyboard punchers.

Editing for Consistency

Editing requires checking for logically consistent responses. The in-house editor must determine if the answers given by a respondent to one question are consistent with those given to other related questions.

Editing for Completeness

In some cases, the respondent may have answered only the second portion of a two-part question. An in-house editor may have to adjust the answers to the following question for completeness.

Item Non-response

Item non-response is a technical term for an unanswered question on an otherwise complete questionnaire. Specific decision rules for handling this problem should be meticulously outlined in the editorial instructions.

Editing Questions Answered out of Order

Another situation an editor may face is the need to rearrange the answers to an open-ended response to a question.

Coding

Coding involves assigning numbers or other symbols to answers so the responses can be grouped into a limited number of classes or categories. Codes allow data to be processed in a computer.

Code Construction

When the question has a fixed-alternative (closed-ended) format, the number of categories requiring codes is determined during the questionnaire design stage. The coding categories should be exhaustive and mutually exclusive.

Code Book

A code book is a document describing the coding procedure and the location of data for variables in a format that computers can use.

Production Coding

Transferring the data from the questionnaire or data collection form after the data have been collected is called production coding.

Data Entries

Use of scanner sheets for data collection may facilitate the entry of the responses directly into the computer without manual keying in the data.

Cleaning Data

The final stage in the coding process is the error-checking and verification or “data cleaning” stage, which is a check to make sure that all codes are legitimate. Accuracy is extremely important when coding data. Errors made when coding or entering data into a computer threaten the validity of measures and cause misleading results.

LESSON:30 DATA TRANSFROMATION

Data Transformation

Data transformation changes data from its original form to a format suitable for analysis to meet research objectives. Researchers often modify scalar data values or create new variables. For instance, birth year data can be transformed into age by subtracting the birth year from the current year. Combining categories, such as merging "strongly agree" and "agree" responses, is another common transformation.

Summative Score = Variable 1 + Variable 2 + Variable 3

This calculation can be accomplished by using simple arithmetic or by programming a computer with a data transformation equation that creates the new variable “summative score.”

Creating new variables through arithmetic or programming is essential, like calculating a summative score from various attitudinal statements. Many scales and indexes measure social phenomena, evolving over time to refine constructs and definitions. Researchers can use existing measures or modify them as needed.

Indexes and Scales

Scales and indexes provide ordinal- or interval-level measures of variables. A scale captures the intensity, direction, level, or potency of a variable on a continuum and can use single or multiple indicators. An index combines several indicators into a single score, often a simple sum, and is used for content or convergent validity. Both scales and indexes enhance reliability, validity, and data reduction.

Unidimensionality

Unidimensionality ensures all items in a scale or index measure a single construct. For example, feminist ideology includes beliefs in social, economic, political, family, and sexual relations, forming a system of beliefs about women's dignity and power.

Index Construction

An index combines items into a single numerical score. For instance, the Consumer Price Index (CPI) measures inflation by totaling the cost of a list of goods and comparing it to previous years. Creating indexes involves ensuring items have face validity and measuring each construct part with at least one indicator.

To show how simple indexes are, answer "yes" or "no" to these questions for four occupations: long-distance truck driver, medical doctor, accountant, and telephone operator. Score 1 for "yes" and 0 for "no."

1. Good salary?
2. Job security?
3. Interesting and challenging work?
4. Good working conditions?
5. Career advancement opportunities?
6. Prestigious?
7. Self-direction and decision freedom?

Total the scores for each occupation to compare.

A college quality index might include six characteristics: student-to-faculty ratio, faculty education, library size, dropout rates, advanced degree attainment, and faculty publications. Combining sub-indexes, like teaching quality, can further elaborate the construct.

Weighting

Index construction may involve weighting items, although unweighted indexes are standard unless there is a reason for assigning different weights.

Scoring and Score Index

Scoring items, like job satisfaction statements, involves assigning values based on agreement levels, with positive and negative statements scored appropriately. A score index ranges from minimum to maximum possible scores, indicating levels of satisfaction. Scores can be used for statistical analysis and hypothesis testing, or categorized for presentation and cross-classification.

LESSON 31 DATA PRESENTATION

Tables and Graphs

Tables and graphs simplify and clarify research data by summarizing and communicating its meaning visually. They can take various forms, from simple computer printouts to elaborate pictographs. Common types of graphic presentations include bar charts, pie charts, curve diagrams, and pictograms, which create strong visual impressions. The advancement of computer technology has made data tabulation and statistical analysis more accessible, with statistical packages like SAS, SPSS, SYSTAT, Epi Info, and MINITAB being commonly used. These packages are user-friendly and provide extensive programs for data analysis.

Results with One Variable

Frequency Distribution

The frequency distribution is a straightforward method to describe the numerical data of one variable. It can be used with nominal, ordinal, interval, or ratio-level data. For example, to summarize the gender data of 400 students, a frequency distribution table can be used:

- **Table 1: Frequency distribution of students**
  - Male: 300 (75%)
  - Female: 100 (25%)
  - Total: 400 (100%)

This data can also be represented graphically through histograms, bar charts, or pie charts. Bar charts are used for discrete variables and can be oriented vertically or horizontally. Histograms are typically upright bar graphs for interval or ratio data.

Interpretation of Data

Interpreting data involves explaining its meaning to draw inferences and formulate conclusions. The interpretation process requires meaningful analysis, often facilitated by statistics.

Descriptive Statistics

Descriptive statistics describe numerical data and can be categorized based on the number of variables involved:

- **Univariate Statistics**: Describe one variable and often use measures of central tendency, including mean, median, and mode. The mode is the most frequent number, the median is the middle point, and the mean is the arithmetic average.
- **Bivariate Statistics**: Describe two variables.
- **Multivariate Statistics**: Describe three or more variables.

Bivariate Tables

The bivariate contingency table is widely used for cross-tabulation, organizing cases based on two variables simultaneously. This method shows how the cases in one variable's category are "contingent upon" the categories of another variable.

Constructing Percentage Tables

Constructing a percentage table involves organizing data professionally. For instance, to explore the relationship between age and attitude towards "women empowerment," the age variable can be recoded into three categories (under 40 years, 40-60 years, and 61+ years). This recoding simplifies the analysis and allows for a clearer presentation of the data. A hypothesis could be that lower age correlates with a more favorable attitude towards "women empowerment." This hypothesis can be tested by cross-tabulating age and attitude variables in a percentage table.

By presenting data in tables and graphs, researchers can make complex data more understandable and draw meaningful conclusions from their analyses.

LESSON 32 THE PARTS OF THE TABLE

- **Parts of the Table**

  - **Numbering and Title**: Each table is numbered and titled with background information and variable names.
  - **Labeling Rows and Columns**: Variables are labeled for rows and columns, categorizing each variable.
  - **Including Totals**: Totals for rows and columns (marginals) are included, representing the univariate frequency distribution.
  - **Cells and Body of the Table**: Each cell corresponds to an intersection of variables, forming the body of the table.
  - **Handling Missing Information**: Missing cases are reported near the table to account for all original cases.

- **Percentaging Tables**: Raw count tables are converted into percentages to analyze bivariate relationships. Percentages can be computed by row, column, or total, depending on the research hypothesis and the causal direction implied.

- **Reading Percentage Tables**: Understanding table titles, variable labels, and computation directions (row or column) is crucial. Comparisons are made across rows for column-percentaged tables and up/down columns for row-percentaged tables.

- **Interpreting Relationships**: Identifying relationships in percentage tables involves looking for patterns. Linear relationships show larger percentages in diagonal cells, while curvilinear relationships exhibit patterns across cells.

- **Statistical Control**: Establishing a causal relationship between variables requires more than just showing an association. Researchers must eliminate alternative explanations that could distort findings. In experimental research, this is achieved through rigorous design to control for

potential confounders. In non-experimental studies, statistical methods are used to identify and measure these factors using control variables.

Multivariate analysis, involving three or more variables, helps researchers determine if a relationship is genuine. For instance, when studying attitudes towards women empowerment, researchers might initially find a correlation with age. However, by controlling for gender, they can assess whether age independently influences attitudes across different gender groups.

Statistical control is essential to ensure that observed relationships are not spurious. It involves comparing relationships before and after introducing control variables to understand their true impact. Methods such as trivariate tables and multiple regression analysis are employed to dissect these relationships and gauge the net effect of independent variables while accounting for potential confounders

.

- **Trivariate Tables**: Researchers aiming to establish causal relationships must address alternative explanations that could invalidate their findings. These alternative explanations are operationalized as control variables, which help mitigate confounding factors.

To assess the influence of these control variables on bivariate relationships, researchers use trivariate or three-variable tables. Unlike bivariate tables that examine two variables, trivariate tables include multiple bivariate analyses corresponding to each category of the control variable. These subsets, called partials, focus on cases with specific values of the control variable.

Trivariate tables have THREE limitations: they can be challenging to interpret with control variables having more than four categories. Additionally, control variables of different measurement levels need to be appropriately grouped to maintain consistency in interpretation. The total number of cases is also crucial as it determines the viability of partial analyses within the trivariate framework.

In constructing trivariate tables, researchers start with a compound frequency distribution (CFD), expanding it into a three-way distribution to incorporate the control variable's categories. This method allows for a comprehensive analysis of relationships while controlling for potential confounders like gender, as illustrated in examples within the research context. The number of cells in the partials equals the number of cells in the bivariate relationship multiplied by the number of categories in the control variables. For example if the control variable has three categories, and a bivariate table has 12 cells, the partials have 3 X 12 = 36 cells. An average of five cases per cell is recommended, so the researcher will need 5 X 36 = 180 cases at minimum

- **Multiple Regression Analysis**: This statistical technique accounts for multiple alternative explanations simultaneously, unlike bivariate or trivariate tables which have limitations in handling more complex relationships.

LESSON 33

EXPERIMENTAL RESEARCH

Experimental Research Experimental research is a methodical approach used primarily in the natural and social sciences to establish causal relationships between variables. It involves manipulating one variable (independent variable) under controlled conditions to observe its effect on another variable (dependent variable), while controlling for other factors that could influence the outcome.

Key Components of Experimental Research:

1. **Treatment or Independent Variable:**
   - The variable that researchers manipulate or control to observe its effect on the dependent variable.
   - Example: In a study on plant growth, different amounts of fertilizer applied (independent variable) to test how they affect plant growth (dependent variable).
2. **Dependent Variable**:
   - The variable that is measured or observed to determine the effect of the independent variable.
   - Example: Plant height, yield, or other growth metrics in the plant growth experiment.
3. **Pretest**:
   - Measurement of the dependent variable before the treatment is applied.
   - Helps establish a baseline against which post-treatment changes can be compared.
4. **Posttest**:
   - Measurement of the dependent variable after the treatment is applied.
   - Allows researchers to assess changes caused by the independent variable.
5. **Experimental Group**:
   - Group of subjects exposed to the experimental treatment (manipulated independent variable).
   - Example: Plants receiving different amounts of fertilizer in the plant growth study.
6. **Control Group**:
   - Group of subjects that does not receive the experimental treatment, used as a baseline for comparison.
   - Helps account for external factors and ensures any observed effects are due to the treatment.
7. **Assignment of Subjects**:
   - Process of assigning participants to experimental and control groups.
   - Random assignment helps minimize bias and ensures groups are comparable.

Experimental research allows researchers to establish causal relationships by systematically manipulating variables under controlled conditions. It provides a rigorous framework for testing hypotheses and drawing conclusions about cause and effect relationships in various fields of study.

Dependent Variable:

- The dependent variable is what researchers measure to assess the impact of changes in the independent variable.
- Example: In a study, turnover rate, absenteeism, or morale might be measured to see how training affects these outcomes.

Pretests and Posttests:

- Researchers often measure the dependent variable before (pretest) and after (posttest) applying the treatment (independent variable).
- This helps establish a baseline and compare changes caused by the treatment.

Experimental and Control Groups:

- Subjects are divided into experimental groups (receive treatment) and control groups (do not receive treatment).
- This setup helps isolate the effects of the independent variable by comparing outcomes between groups.

Assignment of Subjects/Test Units:

- Random assignment ensures subjects are assigned to groups in an unbiased way.
- It helps ensure that groups are similar in characteristics, except for the independent variable being tested.

Three Types of Controls:

1. **Manipulation of the Independent Variable**:

Researchers manipulate or change the independent variable to observe its effects.

- Example: Testing how different levels of lighting affect worker productivity among sewing machine operators.

Experimental research allows researchers to carefully test hypotheses by manipulating variables under controlled conditions. This method helps establish causal relationships and draw meaningful conclusions in various fields of study.

2. **Holding Conditions Constant:**

- When studying cause-and-effect relationships between variables X and Y, other factors (like A) might also influence Y.
- To accurately assess the effect of X, researchers need to control or eliminate the effects of these other factors, known as confounding factors.

3. **Control over the Composition of Groups**:

- Experimental and control groups should be carefully selected or matched to ensure that any differences between them do not affect the results.
- Random selection or matching ensures that groups are comparable, allowing researchers to isolate the effects of the independent variable (X).

Random Assignment:

- Random assignment involves assigning subjects (e.g., individuals or organizations) to different experimental groups purely by chance.
- This method minimizes bias and ensures that groups do not systematically differ from each other, except for the treatment they receive.

Matching:

- Matching involves pairing subjects based on specific characteristics (e.g., age, gender) to create groups that are similar in those aspects.
- While matching can be effective, it is less commonly used than random assignment because it can be challenging to find exact matches for all relevant characteristics.

In experimental research, these methods help researchers control variables and ensure that observed effects are due to the manipulated independent variable rather than other factors. Random assignment is generally preferred for its ability to create comparable groups efficiently.

LESSON 34
EXPERIMENTAL RESEARCH (CONT.)

Steps in Conducting an Experiment:

1. **Formulate a Hypothesis**: Create a clear hypothesis that can be tested through experimental research.
2. **Choose an Experimental Design**: Decide on the structure of the experiment, including the number of groups, how the treatment will be applied, and when measurements will be taken.
3. **Implement the Treatment**: Introduce the independent variable (treatment) in a controlled manner.
4. **Measure the Dependent Variable**: Develop reliable methods to measure the outcome affected by the independent variable.
5. **Pilot Test**: Conduct a trial run to refine the experimental setup and measurement methods.
6. **Select Subjects**: Choose appropriate participants or cases for the experiment.
7. **Random Assignment (if applicable)**: Assign subjects randomly to different experimental groups to minimize bias.
8. **Pretest (if applicable)**: Measure the dependent variable before applying the treatment to establish a baseline.

9. **Apply Treatment**: Administer the treatment to the experimental group(s) while keeping conditions constant for all groups.
10. **Posttest**: Measure the dependent variable again after applying the treatment to assess any changes.
11. **Debrief Subjects**: Inform participants about the experiment's purpose and any deceptive aspects used.
12. **Analyze Data**: Compare results between groups using statistical methods to determine if the hypothesis is supported.

Types of Experimental Designs:

1. **Classical Experimental Design**: Includes random assignment, pretest, posttest, an experimental group receiving treatment, and a control group.
2. **Quasi-Experimental Designs**:
    - **One-shot Case Study Design**: One group receives treatment, and there is only a posttest. No random assignment.
    - **One Group Pretest-posttest Design**: One group receives treatment with both pretest and posttest measures, but lacks a control group and random assignment.
    - **Two Groups Posttest-only Design**: Two groups with random assignment and a posttest measure, but no pretest.
3. **True Experimental Designs**:
    - **Pretest and Posttest Experimental and Control Group Design**: Two groups (experimental and control) with random assignment, pretest, and posttest measures to assess treatment effects.
    - **Solomon's Four Group Design**: Includes two experimental and two control groups, with one set having both pretest and posttest measures for enhanced validity.

Experimental designs vary in complexity and the extent to which they control variables, aiming to establish causal relationships between variables under study.

Solomon's Four Group Design:

- **Experimental Group 1**: Pretest (O1), Treatment (X), Posttest (O2)
- **Control Group 1**: Pretest (O3), Posttest (O4)
- **Experimental Group 2**: No Pretest, Treatment (X), Posttest (O5)
- **Control Group 2**: No Pretest, Posttest (O6)

Interaction Effect:

- When two variables (like smoking and smog) together have a greater effect than the sum of their individual effects.
- Example: If smoking alone has a 20% cancer rate and smog alone has a 10% rate, but together they have a 37% rate, the interaction effect is 7%.
- Interaction Effect = Total effect – (smoking effect + smog effect) = 37 percent = 37 percent - (20 percent + 10 percent) = 37 percent - 30 percent = 7 percen

Pretest Effect:

- The possibility that the act of measuring (pretest) can influence the participants' responses, affecting the experiment's outcomes.
- It can be observed in both experimental and control groups, influencing the interpretation of results.

Other Experimental Designs:

- **Randomized Block Design**: Divides subjects into blocks (homogeneous groups) and then randomly assigns treatments within each block.
- **Latin Square Design**: Ensures each treatment appears exactly once in each row and column of a grid, controlling for variables that could affect the outcome.
- **Natural Group Design**: Observes existing groups without any manipulation.
- **Factorial Design**: Examines multiple independent variables simultaneously to understand their combined effects.

These designs vary in complexity and purpose, aiming to minimize bias and establish clear cause-and-effect relationships in experimental research.

LESSON 35 EXPERIMENTAL RESEARCH (CONT.)

Internal Validity in Experiments:

- **Definition:** Internal validity assesses whether the independent variable (the cause) is solely responsible for changes observed in the dependent variable (the effect), without interference from other variables.

Factors Affecting Internal Validity:

1. **History Effect:**
    - **Description:** External events occurring between measurements that can affect the outcome.
    - **Example:** A product's sales affected by contamination of drinking water, an event outside the researcher's control.
2. **Maturation Effect:**
    - **Description:** Changes over time within subjects due to natural biological or psychological processes.
    - **Example:** Employees' performance improvement attributed to time passing rather than technology introduction.
3. **Testing Effects:**
    - **Description:** The impact of taking a pretest on subjects' responses, influencing their behavior in subsequent tests.
    - **Example:** Subjects sensitized by pretest may alter responses in the posttest, affecting the interpretation of treatment effects.
4. **Instrumentation Effects:**

   - **Description:** Changes in measuring instruments or procedures between pretest and posttest that affect results.
   - **Example:** Scale readings changing due to mechanical issues, leading to inaccurate posttest measurements.
5. **Selection Bias Effect:**
   - **Description:** Non-equivalent group formation due to biased selection of subjects, affecting comparability.
   - **Example:** Experimental group unintentionally containing athletes, while control group comprises artists, biases results in studies on aggressiveness.
6. **Statistical Regression:**
   - **Description:** Extreme scores on initial measurement tending to move towards the average on subsequent measures due to random error.
   - **Example:** High-security prisoners showing less violence after exposure to violent films could be due to statistical regression, not the film's effect.
7. **Mortality:**
   - **Description:** Dropout of subjects during the experiment, leading to unbalanced groups and potentially biased results.
   - **Example:** Salespersons dropping out of a high-pressure supervision experiment may skew results, making high-pressure appear more effective.

These factors must be carefully considered and controlled to enhance the internal validity of experimental research, ensuring that observed effects can be confidently attributed to the independent variable.

Mechanical Loss:

- **Description:** Equipment failure during an experiment can lead to unreliable results.
- **Example:** Video equipment malfunctioning during a study on behavior affects data validity.

Experimenter Expectancy:

- **Description:** Bias introduced when researchers unintentionally communicate expectations to participants.
- **Example:** Nonverbal cues from a researcher affect how male and female subjects report feelings toward disabled individuals.
- **Control:** Double-blind experiments mitigate this by keeping both subjects and experimenters unaware of the treatment details.

External Validity:

- **Description:** The extent to which experimental findings can be generalized to real-world situations.
- **Issue:** Without external validity, findings may lack practical applicability beyond the specific experiment's conditions.

Reactivity: A Threat to External Validity:

- **Description:** Subjects may alter their behavior due to being observed in an experiment, affecting the validity of real-world applicability.
- **Example:** The Hawthorne Effect shows increased productivity due to participants feeling important rather than actual changes in working conditions.
- **Demand Characteristic:** Participants may change behavior based on perceived expectations of experimenters, influencing outcomes.

Ethical Issues in Lab Experiments:

- **Concerns:** Various ethical considerations arise in experimental research.
- **Examples:** Coercing participation, deceiving subjects about research purposes, exposing participants to stress, and not ensuring privacy or debriefing.

Human Subjects Committee:

- **Purpose:** Ethics committees safeguard participant rights in research.
- **Function:** They oversee and ensure ethical conduct in studies, including experimental designs.

These summaries cover the main points of mechanical loss, experimenter expectancy, external validity, reactivity, ethical issues, and the role of ethics committees in experimental research.

LESSON 36 NON-REACTIVE RESEARCH

Experiments and Surveys (Reactive Research):

- **Definition:** Participants are aware they are being studied, influencing their behavior or responses.
- **Purpose:** To test hypotheses and gather data through controlled settings or structured questionnaires.

Non-Reactive Research:

- **Definition:** Participants are unaware they are part of a study, allowing natural behaviors to be observed.
- **Methods:**
    - **Unobtrusive Measures:** Observing behaviors without disrupting them, such as unnoticed observation or indirect indicators.
    - **Examples:** Watching drivers from a distance to see if they stop at traffic lights, noting details like gender, presence of passengers, and stopping behavior.

Varieties of Non-Reactive Observations:

- **Physical Traces:** Using wear and tear on objects (like toys) or accumulation of items (like soft drink cans) to infer behaviors.
- **Archives:** Analyzing public records (e.g., marriage records) or private records (e.g., office paper purchases) for insights.
- **Observations:** Noting external appearances (e.g., clothing choices after a sports team's win) or counting behaviors (e.g., gender differences in stopping at traffic signs).

Recording and Documentation:

- **Logic:** Follows quantitative measurement principles to systematically record and analyze empirical evidence.
- **Application:** Used by both quantitative and qualitative researchers to link conceptual constructs to observable data.

Content Analysis:

- **Definition:** Examines text (written, visual, or spoken) to uncover meanings, themes, or messages.
- **Texts Studied:** Includes books, articles, advertisements, speeches, films, and other forms of communication.
- **Process:** Involves systematic counting and recording to quantitatively describe symbolic content (textual coding).
- **Non-Reactive Nature:** Analyzes text content without interacting with its creators, ensuring objectivity.
- **Benefits:** Provides quantitative insights into text content, allows comparisons across texts, and reveals subtle patterns or biases.

Purpose and Applications:

- **Utility:** Used in fields like literature, history, psychology, and sociology to study communication effects and patterns.
- **Analysis:** Data gathered from content analysis are entered into computers for statistical analysis, similar to experiment or survey data.

In summary, non-reactive research and content analysis offer methodologies to study behaviors and communication without directly influencing participants, providing valuable insights across various disciplines.

Measurement and Coding in Content Analysis:

Careful Measurement:

- **Importance:** Essential for converting ambiguous symbolic communication into precise, objective, quantitative data.
- **Procedure:** Researcher designs and documents coding procedures meticulously to ensure replicability.

- **Example:** Assessing how often elderly characters are negatively stereotyped in TV dramas involves defining stereotypes (e.g., senile, frail) that might inaccurately portray older adults.

Constructs and Coding System:

- **Definition:** Constructs like "negatively stereotypes of the elderly" or "leadership role" are operationalized with specific coding instructions.
- **Application:** Coding rules clarify how to classify content from text based on predefined categories (e.g., upper, middle, lower class in social class analysis).

Structured Observations:

- **Types of Measurement:**
  - **Frequency:** Counting occurrences (e.g., how often elderly characters appear in TV programs).
  - **Direction:** Noting positive or negative tones in content (e.g., friendly vs. nasty traits in character descriptions).
  - **Intensity:** Assessing the strength or significance of traits (e.g., minor vs. major forgetfulness).
  - **Space:** Measuring physical or temporal presence (e.g., duration of a TV character's appearance).

Coding in Content Analysis:

Manifest Coding:

- **Definition:** Coding visible, surface-level content in text, such as counting specific words or actions.
- **Reliability:** Highly reliable but may overlook contextual nuances or multiple meanings of words.
- **Example:** Counting occurrences of specific phrases (e.g., "red") in written text or actions (e.g., shaking hands) in visual media.

Latent Coding:

- **Definition:** Analyzing underlying meanings or themes in text beyond surface content.
- **Challenges:** Less reliable due to interpretation variability among coders, requires training and clear rules.
- **Example:** Identifying themes like romance or vulgarity in a paragraph based on general coding guidelines.

Ensuring Reliability:

- **Training and Consistency:** Researchers train coders to ensure understanding and consistency in applying coding rules.

- **Inter-Coder Reliability:** Checking consistency among coders by having them independently code the same texts and measuring agreement statistically.

Conducting Content Analysis Research:

Question Formulation:

- **Initial Step:** Researchers start with clear research questions focusing on symbolic variables (e.g., portrayal of women in advertisements).

Unit of Analysis:

- **Definition:** Determining the specific amount of text or content (e.g., each advertisement) assigned for coding and analysis.

Sampling:

- **Procedure:** Often involves random sampling of text units (e.g., articles in magazines) to ensure representative data.
- **Example:** Selecting articles from weekly news magazines published between 2001 and 2007 to study portrayals of women.

Variables and Coding Categories:

- **Definition:** Operationalizing variables like "significant leadership role" with clear coding instructions and categories (positive vs. negative portrayals).
- **Example:** Defining leadership qualities (e.g., brilliant, uninspired) to categorize roles in articles featuring women.

Inferences:

- **Limitations:** Content analysis describes content within texts but does not infer creators' intentions or the effects of messages on audiences.

In summary, measurement and coding in content analysis involve systematic approaches to extract quantitative data from textual content, ensuring rigorous analysis and interpretation of symbolic communication.

LESSON 37 USE OF SECONDARY DATA

Lesson 37: Use of Secondary Data

Advantages of Content Analysis

1. **Access to Inaccessible Subjects**

   - Enables research on subjects like ancient civilizations or speeches of historical figures.
2. **Non-Reactivity**
   - Minimal influence from researcher; documents written for other purposes remain unaltered.
3. **Longitudinal Analysis**
   - Suited for studying trends over time, comparing different periods in history.
4. **Use of Sampling**
   - Allows random sampling to study specific aspects across a population of documents.
5. **Large Sample Size**
   - Facilitates more confident generalizations compared to experimental or survey research.
6. **Spontaneity**
   - Captures spontaneous actions or feelings recorded at the moment, like personal diaries.
7. **Confessions**
   - Documents like diaries or autobiographies may reveal personal insights not shared otherwise.
8. **Relatively Low Cost**
   - Cost-effective compared to large-scale surveys or experiments, especially when documents are centralized.
9. **High Quality**
   - Offers well-written insights, such as expert commentaries in newspapers.

Disadvantages of Content Analysis

1. **Bias**
   - Documents may be biased based on their original purpose or author's intent.
2. **Selective Survival**
   - Common documents may not survive over time or be preserved for research.
3. **Incompleteness**
   - Often lack detailed context or assume prior knowledge not accessible to researchers.
4. **Lack of Availability**
   - Some areas of study lack accessible documents, or documents are classified or destroyed.
5. **Sampling Bias**
   - Documents may disproportionately represent certain demographics or educational levels.
6. **Limited to Verbal Behavior**
   - Provides insights only into verbal expressions, not nonverbal behavior.
7. **Lack of Standardized Format**
   - Variability in document formats complicates comparison across time or sources.
8. **Coding Difficulties**
   - Difficulty in quantifying qualitative data due to varied content and length of documents.

9. **Adjustment for Comparability Over Time**
    - Changes in external factors like language or measurement units require careful adjustment.

Use of Secondary Data: Existing Statistics/Documents

1. **Secondary Data**
    - Information collected by others, typically historical and already assembled for various purposes.
2. **Selecting Topic for Secondary Analysis**
    - Researchers reinterpret existing data to address new research questions, leveraging available information creatively.
3. **Sources of Existing Statistics**
    - Includes government reports, international agencies, and private sources like libraries.

Reliability and Validity

1. **Validity**
    - Concerns arise when researcher definitions differ from official definitions used in data collection.
2. **Reliability**
    - Challenges in consistent measurement over time due to changes in data collection methods or definitions.

Inferences from Non-Reactive Data

- **Limitations in Causal Inference**
    - Difficulties in establishing causality or temporal order with non-reactive data; focuses on correlation among variables.

This summary highlights the advantages, disadvantages, and practical considerations of using content analysis and secondary data in research contexts.

LESSON 38

OBSERVATION STUDIES/FIELD RESEARCH

Observation Studies in Qualitative Research Observation studies are part of qualitative research, which focuses on understanding social life and how people construct meanings in natural settings.

Qualitative vs. Quantitative Research

- **Qualitative Research**: Produces soft data like impressions, words, and photos. It uses an interpretive approach and follows a non-linear research process.

- **Quantitative Research**: Produces hard data in numbers, following a positivist approach. It emphasizes precise measurement, hypothesis testing, and establishing causality. It typically follows a linear research process.

Participant vs. Non-Participant Observation

- **Participant Observation**: The researcher directly interacts with and observes people in natural settings to understand their perspectives.
- **Non-Participant Observation**: Researchers observe people without interacting directly, using methods like mirrors or CCTV.

Field Research Field research involves studying people in natural settings and includes methodologies like ethnography and ethnomethodology.

Ethnography and Ethnomethodology

- **Ethnography**: Describes cultures from the native point of view, understanding different ways of life.
- **Ethnomethodology**: Examines how people create and interpret reality through detailed study of social interactions.

Logic of Field Research Field research emphasizes naturalistic observation, grasping social meanings, and maintaining both insider and outsider perspectives.

Roles and Challenges of Field Researchers Field researchers observe daily activities, engage with subjects, maintain analytic distance, and cope with personal and ethical challenges.

Steps in Field Research

1. **Prepare Yourself**: Read relevant literature and focus on the research goals.
2. **Site Selection**: Choose a suitable location and obtain permission to conduct research.
3. **Entering the Field**: Establish relationships with the community members and integrate into the social environment.
4. **Adopting a Social Role**: Understand social dynamics, learn norms, and build rapport with participants.
5. **Data Collection**: Observe, listen attentively, and document data meticulously.
6. **Data Analysis**: Start analyzing collected data, develop hypotheses, and evaluate findings.
7. **Focused Study**: Concentrate on specific aspects using theoretical sampling to deepen understanding.
8. **Field Interviews**: Conduct field interviews with member informant

9. **Disengagement**: Disengage and physically leave the setting.
10. **Analysis and Reporting**: Complete the analysis and write the report.

LESSON 39 OBSERVATION STUDIES (CONTD.)

1. **Preparation and Defocusing**: Before starting field research, it's crucial to immerse oneself in relevant literature to understand concepts, methods, and potential challenges. Emotionally preparing involves reading diverse sources like diaries and novels to gain familiarity. The researcher should clear their mind of preconceptions and adopt a broad perspective beyond their professional role.
2. **Selecting a Site and Gaining Access**: Choosing a field site involves considering social contexts where activities occur across various physical locations. Accessing these sites can pose challenges, ranging from legal barriers to negotiating permissions with gatekeepers who control entry. Researchers navigate an "access ladder," starting as outsiders seeking public information and potentially progressing to deeper levels of observation and interaction.
3. **Entering the Field and Establishing Social Relations**: Researchers enter the field by presenting themselves in a culturally acceptable manner, often learning the local language. They build rapport through friendly interactions, aiming to move from understanding to empathy by sharing experiences with community members.
4. **Adopting a Social Role and Learning the Ropes**: Researchers may adopt existing roles or create new ones to gain access and navigate the field site. They learn to manage personal stress, normalize their presence, and act as an "acceptable incompetent," showing interest in learning rather than expertise.
5. **Observing and Collecting Data**: Field researchers meticulously observe using all senses to gather high-quality data. They rely on detailed field notes containing various media (maps, photographs, recordings) and ensure notes are thorough, organized, and confidential.
6. **Analyzing Data and Generating Hypotheses**: Researchers analyze data in the field to explore research questions and develop working hypotheses based on initial findings. They refine hypotheses through ongoing field research.
7. **Focusing on Specific Aspects and Theoretical Sampling**: Researchers narrow their focus to specific issues after gaining a general understanding. They employ theoretical sampling, guided by evolving theories, to select observations across different times, situations, locations, and people.
8. **Field interviews in field research are characterized by their unstructured, non-directive nature, aimed at deep exploration and understanding. Here's a summary:** Field researchers conduct unstructured, in-depth interviews that differ significantly from formal survey interviews. These interviews involve asking open-ended questions, actively listening, expressing genuine interest, and recording detailed responses. The process is a collaborative effort between the researcher and participants, where members actively contribute insights and feelings, shaping the conversation to reveal subjective meanings. The interviewer's approach—listening attentively, encouraging discussion, and adapting to the participant's cues—is crucial in eliciting rich, contextually embedded data. Field interviews occur in natural settings, allowing for informal,

spontaneous interactions that are essential for capturing nuanced perspectives and experiences.

A comparison of the field research interview and a survey interview could be as below:

Survey Interview	Field Interview
1. It has clear beginning and end.	1. The beginning and end are not clear. The interview can be picked up later.
2. The same standard questions are asked of all respondents in the same sequence.	2. The questions and the order in which they are asked are tailored to specific people and situations.
3. The interviewer appears neutral at all times.	3. The interviewer shows interest in responses, encourages elaboration.
4. The interviewer asks questions, and the respondent answers.	4. It is like a friendly conversational exchange, but with more interviewer questions.
5. It is almost always with one respondent alone.	5. It can occur in group settings or with others in the area, but this varies.
6. It has a professional tone and businesslike focus; diversions are ignored.	6. It is interspersed with jokes, asides, stories, diversions, and anecdotes, which are recorded.
7. Closed-ended questions are common, with rare probes.	7. Open-ended questions are common, and probes are frequent.
8. The interviewer alone controls the pace and direction of the interview.	8. The interviewer and member jointly control the pace and direction of the interview.
9. The social context in which the interview occurs is assumed to make little difference.	9. The social context of the interview is noted and seen as important for interpreting the meaning of responses.
10. The interviewer attempts to mold the communication pattern into a standard framework.	10. The interviewer adjusts to the member's norms and language usages.

9. Disengage and physically leave the setting:

When fieldwork concludes, it may end naturally with theory-building completion or require a deliberate decision to terminate relations. Disengagement can be emotionally taxing for both researchers and participants. The exit process varies based on involvement intensity and duration, ranging from abrupt departure to gradual withdrawal over weeks. Researchers must decide how and when to inform participants, aiming to maintain positive relationships by respecting local norms.

10. **Complete the analysis and write the report**. After disengaging from the field setting the researcher writes the report. The researcher may share the written report with the members observed to verify the accuracy and get their approval of its portrayal in print. It

may help in determining the validity of the findings. However, it may not be possible to share the findings with marginal groups like addicts, and some deviant groups.

Ethical Dilemmas of Field research

Field research involves deep personal involvement in the lives of others, which raises significant ethical dilemmas. These dilemmas often emerge unexpectedly during fieldwork and include:

1. **Deception**: Researchers may face ethical issues related to covert research, assuming false identities, or misleading participants. Covert research is debated for its necessity versus its potential to undermine trust.
2. **Confidentiality**: Researchers are morally obligated to maintain the confidentiality of sensitive information obtained during fieldwork, including disguising members' identities in field notes.
3. **Involvement with Deviants**: Researchers studying deviant behaviors must navigate dilemmas of trust and involvement in illegal activities while maintaining ethical boundaries and personal moral standards.
4. **Power Dynamics**: Researchers often study marginalized groups, but accessing powerful elites can be challenging due to gatekeeping. This can lead to criticisms of bias or neglecting powerful perspectives.
5. **Publishing Field Reports**: Ethical dilemmas arise in deciding what information to publish. Balancing the right to privacy with the public's right to know is crucial. Researchers may seek member verification of reports but must respect privacy, especially for marginalized groups.

These dilemmas highlight the complex ethical terrain that field researchers navigate, requiring careful consideration of moral responsibilities and research integrity.

LESSON 40

HISTORICAL COMPARATIVE RESEARCH

History and Historiography

History encompasses past events, while historiography refers to the methods used to research and analyze historical evidence.

Historical-Comparative Research

This method focuses on historical time and cross-cultural variations, exploring societal changes and why certain social arrangements differ across societies.

Major Questions

Historical-comparative research tackles significant inquiries like the causes of societal revolutions in various countries, the evolution of major social institutions over centuries, and shifts in societal values.

Suitability and Applications

It is ideal for examining the societal factors leading to specific outcomes, comparing entire social systems across cultures, and studying long-term societal changes.

Conceptualization and Theory Building

By examining historical events and diverse cultural contexts, researchers can develop new concepts and broaden perspectives grounded in specific historical and cultural contexts.

Focus of Historical-Comparative Research

This research method traces the development of social forms over time and compares these forms across different cultures to identify commonalities and unique characteristics.

Scientific Approach

Historical-comparative research uses a variety of techniques including surveys of historical events through document analysis, communication studies, and interviews to gather longitudinal and cross-cultural data.

Examples

Researchers have examined the historical progression of social forms from simple to complex societies, and the evolution of economic systems from primitive to capitalistic, with insights drawn from historical evidence.

Logic of Historical-Comparative Research

This section discusses methodological approaches within historical-comparative research, distinguishing quantitative and qualitative methods and their applications in studying historical and comparative issues.

Unique Features

Historical-comparative research distinguishes itself by interpreting limited and indirect evidence, focusing on contingent, combinational explanations rather than deterministic causality.

Similarities to Field Research

Similarities with field research include recognizing the researcher's perspective, immersion in diverse data sources, and the use of grounded theory to understand social reality through actions and processes over time.

Economic forms and ideas;

Karl Marx proposed economic determinism, asserting that economic factors predominantly shaped society. Conversely, Max Weber argued against this view, contending that non-economic ideas could influence economic forms. In his study of the sociology of religion, Weber examined whether religious institutions dictated social behavior rather than merely reflecting economic conditions. His work, notably "The Protestant Ethic and the Spirit of Capitalism," explored how John Calvin's theology, emphasizing predestination and worldly "callings," spurred Calvinists to reinvest wealth, thereby stimulating capitalism. Weber's analysis contrasted this with Judaism, and the religions of China and India, where he found no support for capital accumulation, reinforcing his thesis on Protestantism's role in capitalism's emergence.

Historical-comparative research encompasses diverse approaches and methodologies, leading to ongoing debates about its distinctiveness and methods. Here’s a summary based on the key points:

1. **Definitions and Approaches**:
    - Researchers use terms like historical, comparative, or historical-comparative, often with varying meanings.
    - Debate exists on whether there is a distinct historical-comparative method or if it's merely social research examining past or multiple societies.
2. **Quantitative Approach**:
    - **Positivist Perspective**: Uses quantitative methods to measure variables, test hypotheses, and discover generalizable laws across time and societies.
    - Applies quantitative research techniques with minor adjustments to study historical or comparative issues.
3. **Qualitative Approach**:
    - Emphasizes understanding and interpreting historical data.
    - Uses qualitative methods like case studies to delve into social meanings and contexts, aiming for deep understanding rather than generalizable laws.
4. **Methodological Practices**:
    - Includes time-series data analysis to monitor changes over time (e.g., population, crime rates).
    - Faces challenges with varying definitions of concepts over time and across societies, impacting comparability.
5. **Theoretical Frameworks**:
    - Draws on Max Weber's concept of "verstehen" to interpret behaviors and actions of past societies.
    - Utilizes ideal types and thick description to construct theoretical models and recreate historical realities.

In essence, historical-comparative research spans a spectrum from quantitative positivist approaches aiming at generalizability to qualitative methods focusing on in-depth understanding and context-specific insights. The choice of approach depends on the research goals and the nature of the historical or comparative inquiry.

A Distinct Historical-Comparative Approach:

A distinct historical-comparative research method avoids the extremes of positivist and interpretive approaches. It blends sensitivity to historical or cultural contexts with theoretical generalization, often using both quantitative and qualitative data. Its goals align closely with field research rather than traditional positivist approaches.

Similarities to Field Research:

1. Both recognize the researcher's unavoidable perspective and involve interpretation.
2. They immerse in diverse data to understand events and people, capturing subjective meanings.
3. Grounded theory emerges during data collection, developing concepts through dialogue with data.
4. Both involve translation of meaning systems for readers.
5. Emphasize action, process, and time in constructing social reality.
6. Limit generalizations and theory due to incomplete, provisional historical and cross-cultural knowledge.

Unique Features of H-C Research:

1. Relies on limited, often indirect evidence like historical documents.
2. Requires interpretation of evidence, acknowledging diverse meanings.
3. Involves reconstruction of past or distant cultures with potential distortion.
4. Avoids deterministic causality, preferring combinational explanations.
5. Shifts between specific and generalized contexts for comparative analysis across cultural or historical units.

These distinctions make historical-comparative research a nuanced approach bridging detailed context with broader theoretical insights.

LESSON 41

HISTORICAL-COMPARATIVE RESEARCH (CONTD.)

Conducting Historical-Comparative Research: A Guideline

1. **Introduction to Historical-Comparative Research**

- Historical-comparative research combines sensitivity to historical or cultural contexts with theoretical generalization.
- It integrates qualitative and quantitative data for comprehensive analysis.

2. **Similarities to Field Research**
   - Both involve interpretation and recognize the researcher's perspective as integral to the study.
   - They focus on diverse data to capture subjective experiences and social meanings.
   - Grounded theory emerges during data collection without fixed hypotheses.
   - They emphasize action, process, and time as essential in constructing social reality.
   - Generalizations are limited and provisional, avoiding fixed laws.
3. **Unique Features of Historical-Comparative Research**
   - Evidence is often limited and indirect, requiring interpretation.
   - Researchers reconstruct the past or other cultures with awareness of contextual biases.
   - Causal explanations are contingent rather than deterministic.
   - Researchers shift between specific and generalized contexts for comparative analysis.
4. **Guidelines for Historical-Comparative Research**
   - **Conceptualizing the Object of Inquiry**
     - Familiarize with the setting and develop preliminary concepts.
     - Orient through reading and formulate specific research questions.
   - **Locating Evidence**
     - Conduct extensive bibliographic research and utilize specialized libraries.
     - Master the literature and evaluate relevance and accuracy of gathered evidence.
   - **Evaluating Quality of Evidence**
     - Assess relevance to research questions and potential biases in interpretation.
     - Critically evaluate the accuracy and authenticity of historical sources.
   - **Organizing Evidence**
     - Systematically organize data and develop initial themes or generalizations.
     - Use theoretical insights to guide data organization and formulate new questions.
   - **Synthesizing**
     - Refine concepts and develop a general explanatory model based on synthesized evidence.
     - Identify patterns across time or units and formulate plausible explanations.
   - **Writing a Report**
     - Integrate evidence, concepts, and synthesis into a coherent research report.
     - Craft the report carefully to communicate findings effectively.
5. **Types of Historical Evidence**
   - Primary sources (e.g., letters, diaries) provide direct accounts from the past.
   - Secondary sources (e.g., books, articles) interpret and analyze historical events.
   - Running records and recollections (e.g., demographic statistics, oral histories) offer additional perspectives.

6. **Problems in Comparative Research**
    - Issues of equivalence challenge comparisons across different historical eras or cultures.
    - Ethical considerations influence methodology and interpretation, though less directly than in other research types.

WANIA HAREEM

Follow this link to join my WhatsApp community:

https://chat.whatsapp.com/GnlgNXt89RPCDqheuikAcB